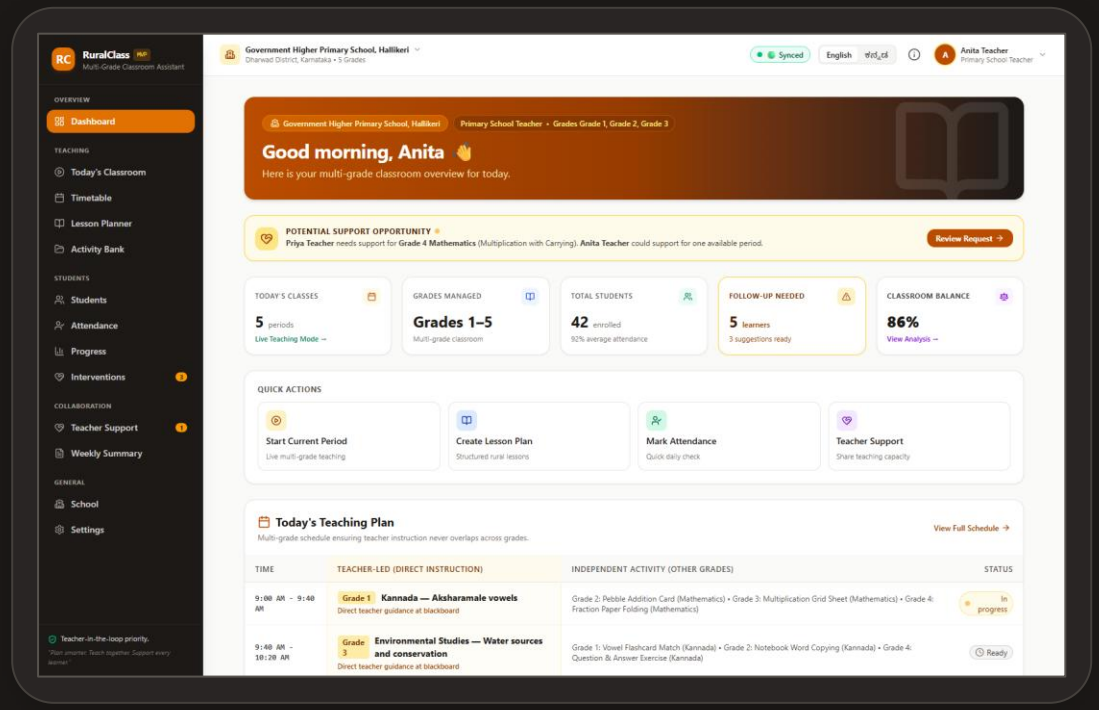
RC **RuralClass**

HACKATHON PITCH · 2026

One teacher. Five grades. One classroom.

RuralClass turns multi-grade chaos into a plan the teacher can actually run.



13 workflows | **13/13 tests** | **EN / ಕನ್ನಡ**

A working prototype — not a slideware promise.

Shreesha N. · github.com/ShreeshaN2006/RuralClass

One adult. Five grades. Six periods. No margin.

Multi-grade teaching is not an edge case in rural India. It is the operating model.

52.1%
of government primary schools have fewer than 60 students

1.04 lakh
single-teacher schools across India

~7%
of all Indian schools run on one teacher

01
The timetable puzzle
Who gets direct instruction — and what do the other four grades do that is actually productive?

02
The planning tax
Five grades means five lesson plans, usually written after the school day ends.

03
The invisible gap
Assessment lives in notebooks; follow-up depends on memory and spare time.

Sources: ASER 2024 National Findings; UDISE+ 2024-25, Ministry of Education, Government of India.

The missing layer isn't content. It's coordination.

Teachers already have textbooks, worksheets and years of practice. What they don't have is a runnable answer to three questions — every single period.

Which grade gets me?

Direct instruction, without splitting attention.

What do the others do?

Independent, group or peer work that needs no screen.

Who needs help next?

A learning gap with a reason and a next step.

RuralClass turns those three questions into one workflow.

One workspace for the teaching day.

Plan the week, run the period, support the learner — without re-entering the same data.

01 Plan

Build a week that fits one room.

- Conflict-free multi-grade timetable
- Timed lesson plans for every grade
- Low-resource activity bank

02 Teach

Run today without improvising.

- Today's Classroom console
- Quick assessments in two minutes
- Attendance in one pass

03 Support

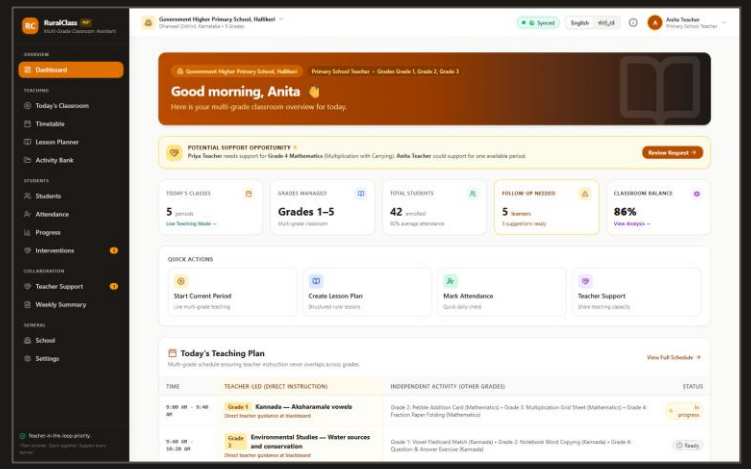
Turn observations into next steps.

- Learner progress at a glance
- Explainable intervention options
- Printable parent summary

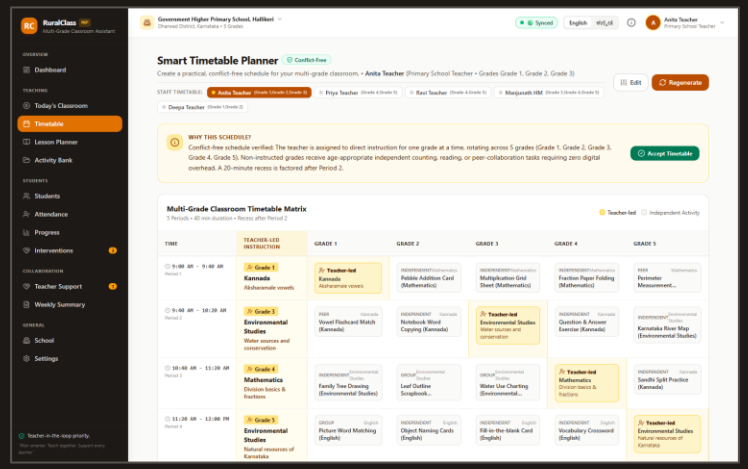
Teacher-in-the-loop · Explainable by default · Bilingual by design

A working prototype, not a mockup.

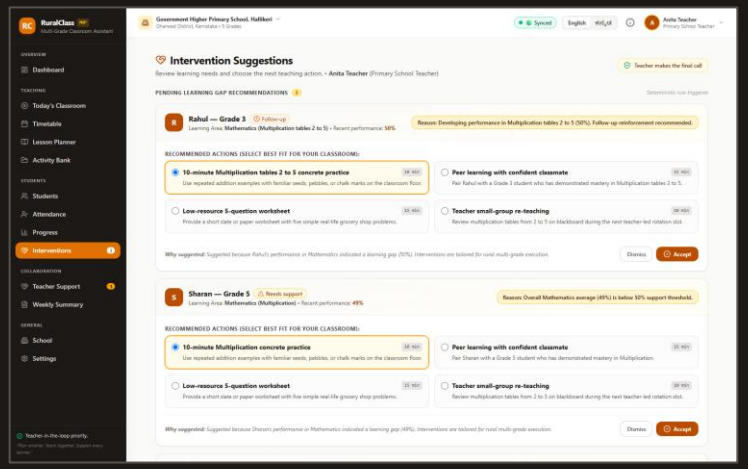
13 teacher workflows, 11 demo accounts and 102 learners — all running in the browser.



Dashboard · the day at a glance



Timetable · conflict-free by construction



Interventions · the why and the next step

Lesson planner · Today's Classroom · Students · Progress · Attendance · Weekly summary · Activity bank · Teacher support

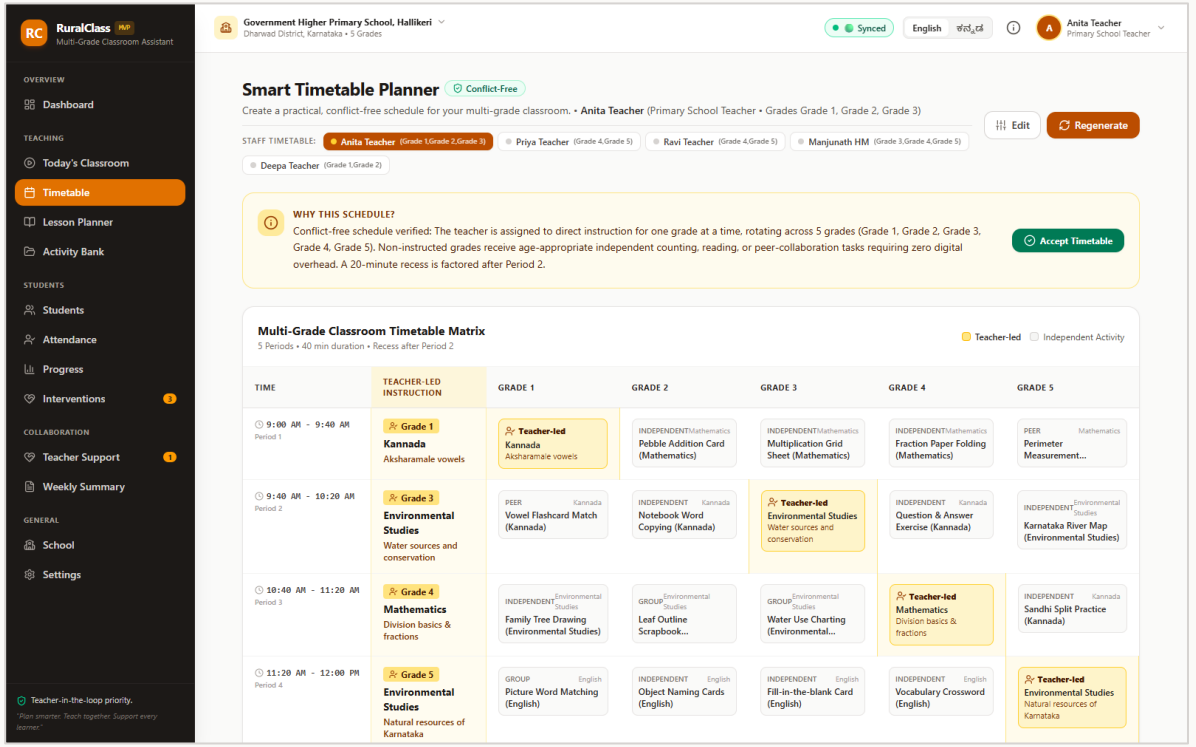
Demo access: teacher anita · password demo123

No API keys. No server calls. Static build.

Conflict-free by construction.

A deterministic scheduler with one hard invariant: exactly one grade gets the teacher in every period.

01	**Rotate**	Pick the next grade, skipping the previous period's grade.
02	**Assign**	Give that grade direct instruction and a subject.
03	**Fill**	Give every other grade independent, group or peer work.
04	**Explain**	Add the recess, format the day and show the reasoning.



Every period: one Teacher-led block, four meaningful independent stations.

✓ **13/13 tests pass — including the exactly-one-teacher-led-grade invariant.**

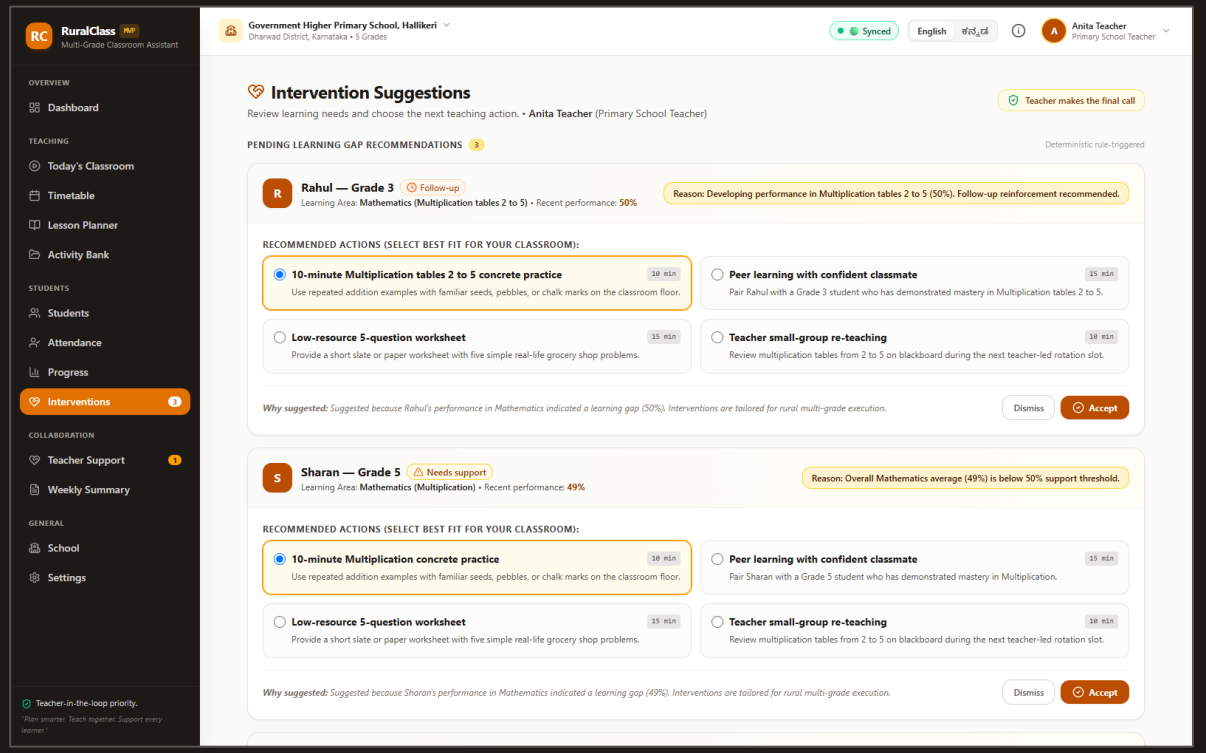
Support that shows its work.

Deterministic rules, visible evidence, four low-resource options. The teacher decides.

2 consecutive scores < 50%	Needs support
Latest score < 50%	Needs support
Latest score 50–59%	Follow-up
No assessments + subject avg < 50%	Needs support

Every suggestion: four concrete options, bilingual, sized for 10–15 minutes.

No opaque model. No hidden score. Teacher-in-the-loop.



The rule, the evidence and the next step are never hidden.

Built for the room, not the demo.

The constraints are the design brief: one shared tablet, intermittent power, no printer, two languages.

EN / ಕನ್ನಡ

Bilingual by default

Every core workflow, including generated teacher copy.

LOW-RESOURCE

No printer required

Slates, seeds, stones and notebook work, not worksheets.

LOCAL-FIRST

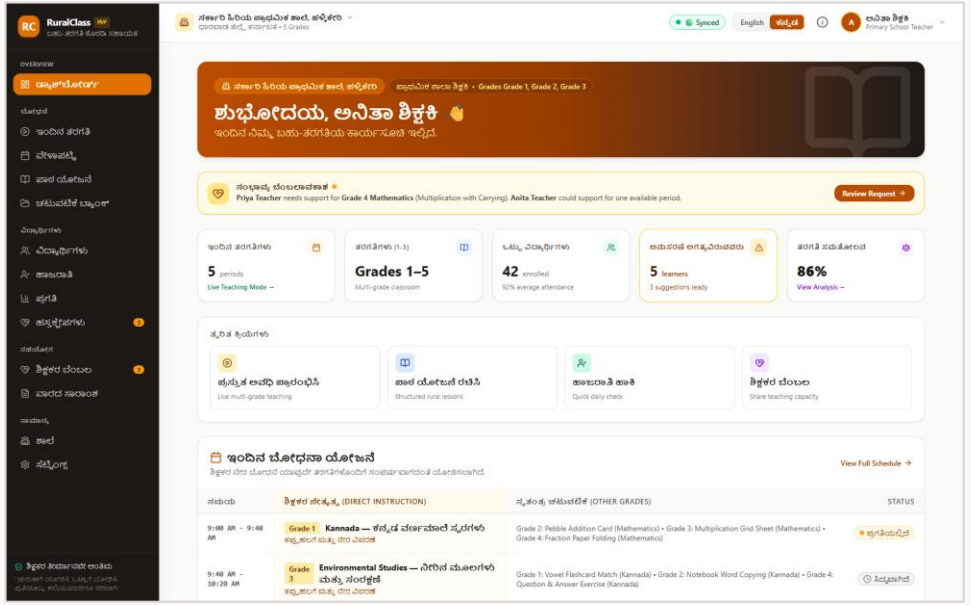
No server in the lesson

Session stays on the device; sync is a roadmap item, not a dependency.

PRIVACY

Minimum data

No Aadhaar, phone numbers or sensitive learner data.



The same dashboard in Kannada — not a translated afterthought.

"Kannada is a first-class interface, not a language toggle."

What changes for the teacher.

Not more data entry. Fewer decisions, made once and reused all day.

	WITHOUT RURALCLASS	WITH RURALCLASS
Timetable planning	**Before** Six periods pieced together in a notebook	**After** A conflict-free schedule in seconds
Lesson preparation	**Before** Five separate plans after school	**After** One plan, plus activities for every other grade
Learner support	**Before** Gaps noticed late, follow-up ad hoc	**After** A reason, four options and a tracked follow-up

4-week pilot scorecard: planning time · interventions completed · attendance follow-up · syllabus coverage

No impact figures claimed yet — these are the metrics we will validate with partner schools.

Small surface area. Testable core.

A static React app with pure decision engines — easy to audit, easy to run on a school device.

React 19 + TypeScript UI

13 views · Tailwind CSS · no component-library lock-in



AppContext

single source of truth for the teacher's day

Timetable

scheduler

Intervention

rule engine

Lesson

planner

Seeded demo data · localStorage session

No backend, no database, no secrets in the browser.

13/13

engine tests passing

0

servers or API keys

13

teacher workflows

EN/KN

full interface

From prototype to a school-ready tool.

The next work is persistence, real accounts and a pilot — not more screens.

NOW

Prototype

- 13 teacher workflows
- Deterministic engines
- Bilingual interface
- 13 passing tests

NEXT

Pilot

- IndexedDB persistence
- School accounts + sync
- 4-week pilot, 3 schools
- Measure planning time & follow-through

THEN

Scale

- Coordinator dashboard
- Print / PDF packs
- More languages
- Accessibility + teacher training

WE ARE HERE · WORKING PROTOTYPE

Indicative timeline; pilot scope will be agreed with partner schools.

R THE ASK

Give every multi-grade teacher a plan they can run.

RuralClass is built, tested and ready for a classroom pilot.

WHAT WE'RE LOOKING FOR

- 3 pilot schools
- Teacher feedback
- Mentorship

Thank you.

Shreesha N. · github.com/ShreeshaN2006/RuralClass

TRY THE PROTOTYPE



github.com/ShreeshaN2006/RuralClass

Demo password: demo123

Run locally with npm install && npm run dev

No sign-up. No API key. No server.